

Shanmugaraja Pandiyan

Software Developer · Machine Learning and Perception for Autonomous Systems

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PROFILE

M.Sc. graduate in Artificial Intelligence and Robotics with a strong foundation in computer vision, deep learning and autonomous systems, and three years of professional software engineering experience at Bosch before returning to study. I build perception pipelines end to end, from sensor calibration and data collection through model development to validation against reference measurements, working in Python, C++ and PyTorch on Linux with ROS 2, Git and CI.

My experience spans **both off road and on road autonomy**: my thesis at IAV develops perception for autonomous agricultural machines, and I have built and calibrated a complete mobile robot sensor stack of GPS, IMU, LiDAR and camera, alongside earlier work on production vehicle software. I am motivated by the point where research meets deployment, and I am looking for an engineering role where I can help move autonomous systems from the lab into real use.

TECHNICAL SKILLS

Languages	Python, C++, C, Bash, SQL
Machine learning	PyTorch, TensorFlow, CNNs, Vision Transformers, object detection (YOLO), semantic and instance segmentation (UNet, DeepLab), behavioural cloning / imitation learning , transfer learning, fine tuning, hyperparameter optimisation, ablation studies, benchmarking
Perception & sensor fusion	OpenCV, camera intrinsic and extrinsic calibration , multi modal sensor calibration, LiDAR and camera fusion, GNSS and IMU fusion (EKF), multi sensor time synchronisation, depth estimation, optical flow, feature matching, multi view geometry
SLAM & localisation	Visual odometry , visual inertial odometry , visual SLAM , GPS/UTM pose estimation, map based localisation, trajectory benchmarking (ATE/RPE, Umeyama alignment)
Environment modelling	Occupancy grid mapping , bird's eye view mapping, inverse perspective mapping, freespace and terrain analysis, point cloud processing (Open3D, PyTorch3D), 3D reconstruction (COLMAP, structure from motion, 3D Gaussian splatting), mesh reconstruction
Robotics & planning	ROS 2 (Humble, Galactic, Jazzy), TF2, Nav2, MoveIt 2, RViz, Foxglove, motion planning (A*, RRT*), PID and behaviour based control, Gazebo, NVIDIA Isaac Sim, CARLA
Embedded & real time	Production embedded C on automotive ECUs , real time systems, AUTOSAR COM and DIAG stacks, CAN, deterministic timing, tight memory budgets, hardware and software in the loop benches
Tooling & process	Linux , Git , GitLab , CI/CD , Docker, CMake, Azure DevOps, JIRA, DOORS, agile / SCRUM, automotive functional safety process
Foundation models	CLIP, SAM2, DINO, vision language models, vision language action models, zero shot perception, automated annotation

EXPERIENCE

2025 to 2026 Berlin, Germany	M.Sc. Thesis Engineer · Perception for Autonomous Machines IAV GmbH <i>Camera to simulation perception and reconstruction pipeline for autonomous agricultural and off road machines</i> - Built an end to end perception pipeline converting monocular video and GNSS into metric, simulation ready environments with bird's eye view maps, terrain primitives and object trajectories. - Developed confidence weighted multi view occupancy mapping to improve terrain consistency and detection of vertical structure, directly analogous to occupancy grid generation for freespace and collision avoidance. - Implemented calibration driven range estimation and corrected systematic yaw error in the monocular visual odometry from roughly 71° to about 1° median, making the localisation output usable downstream. - Built a metric GNSS/UTM pose pipeline and benchmarked trajectories against RTK GNSS using Umeyama alignment, treating validation as part of the work rather than an afterthought. - Developed training free vegetation segmentation and automated terrain and object extraction, and produced georeferenced, dSPACE compatible outputs for simulation and scenario testing. Stack: Python, PyTorch, OpenCV, Open3D, Linux, Git, RTK GNSS, dSPACE ModelDesk, ASAM OpenDRIVE
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2021 to 2023
India
(3 years)

Automotive Software Developer • Embedded Systems
Bosch Global Software Technologies
Production AUTOSAR ECU software for braking and vehicle stability systems (ESP, ABS, IPB)

- Developed **production embedded C software** for real time automotive control units, including AUTOSAR COM and DIAG stack modules delivering reliable CAN communication and OTA diagnostics.
- Worked to hard real time and memory constraints on resource limited hardware, and resolved ECU integration faults across distributed communication layers.
- Automated **software and hardware in the loop test pipelines in Python**, improving validation throughput for real time embedded systems.
- Owned integration, documentation and release workflows under automotive functional safety process, working in an agile team with requirements traceability in DOORS and CI driven delivery.

Stack: C, AUTOSAR, CAN, Python, SiL/HiL, Git, Azure DevOps, DOORS, JIRA

SELECTED PROJECTS

Behavioural cloning	Vision Language Action Mobile Manipulation • Imitation Learning, ROS 2, Isaac Sim • Built a mobile manipulation robot that learns from teleoperated human demonstrations, mapping camera and joint state observations directly to actions rather than following hand written trajectories. • Collected and curated demonstration data in Gazebo and NVIDIA Isaac Sim, implemented ROS 2 nodes for trajectory execution and motion planning from the learned policy, and evaluated task success and error.
Sensor integration	Autonomous Mobile Robot • GPS, IMU, LiDAR and Camera Integration • Built an autonomous mobile robot from the platform up, mounting, wiring and integrating all four sensors on ROS 2, calibrating extrinsics and the TF2 frame tree, and time synchronising the streams. • Fused GNSS and IMU for pose estimation and combined LiDAR and camera for obstacle perception, then validated navigation behaviour on the real machine.
Perception + control	Autonomous Driving Perception and Control • CARLA + ROS 2 • Bridged CARLA with ROS 2 to process synchronised camera, LiDAR, IMU and odometry streams for real time perception, and implemented longitudinal and lateral controllers, closing the perception to control loop across dynamic urban scenarios.
Zero shot perception	Automated Annotation • CLIP + SAM2 • Combined vision language embeddings with promptable segmentation to label unseen object classes without task specific retraining, cutting manual annotation effort and evaluating generalisation across agricultural and driving scenes.
3D reconstruction	High Fidelity 3D Reconstruction • Gaussian Splatting and Photogrammetry • Ran a full structure from motion and 3D Gaussian splatting pipeline (COLMAP, pose recovery, splat optimisation, holdout evaluation) to reconstruct real scenes into simulation ready 3D environments.

EDUCATION

Oct 2024 to 2026	M.Sc. Artificial Intelligence and Robotics • Berlin School of Business and Innovation (BSBI), Germany
2017 to 2021	B.E. Electrical and Electronics Engineering • Anna University, Chennai

LANGUAGES

Languages	English (fluent) • German (A2, actively improving) • Tamil (native)
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